P510/1
PHYSICS
(Theory)
PAPER 1
July/August, 2026
3 hours



LUGAZI DIOCESE MOCK EXAMINATIONS BOARD

Uganda Advanced Certificate of Education

PHYSICS THEORY

Paper 1

3 HOURS

INSTRUCTIONS TO CANDIDATES:

The paper has **Six Items**. *Attempt* **ONLY FOUR** *questions.*

There are **TWO** *sections; Section* **A** *and section* **B**.

Section **A** *is* **COMPULSORY** *and Section B has two parts (i) and (ii).*

Attempt **NOT MORE THAN ONE ITEM** *from each of the Parts (i) and (ii).*

Start each Item on a fresh page.

Use the provided data and support materials to guide your solutions.

Clearly show all calculations and assumptions.

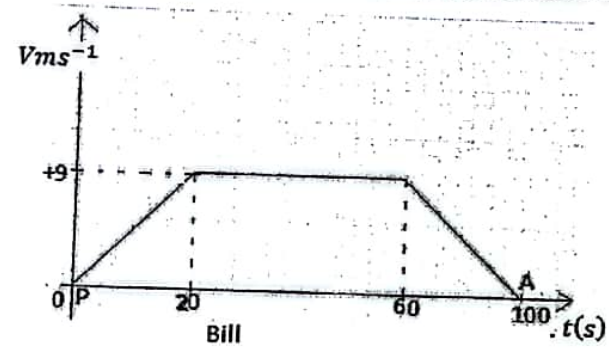
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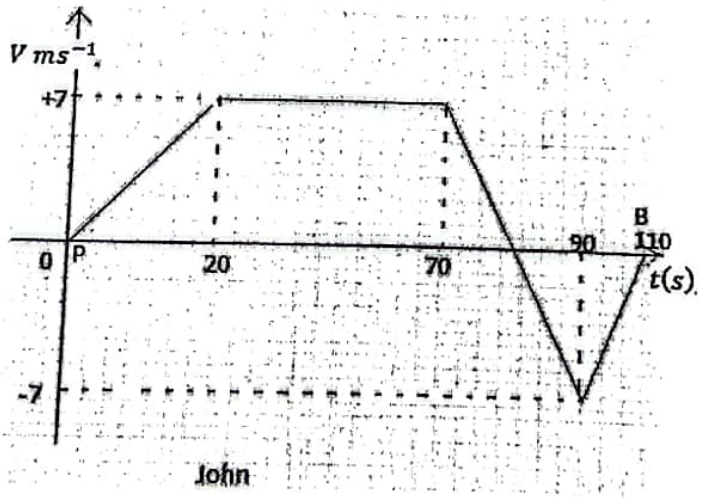
SECTION A (COMPULSORY)

ITEM 1

Athletics in Ugandan schools are utilized to promote holistic education, competition management, physical fitness, and talent identification to foster life skills, social cohesion, and health. In one of the secondary schools, they school coach subjected trial tests to identify various athletes who can represent the school effectively. In running, two boys, Bill and John, started from rest at the same place **P** and each ran respectively to one of the two positions marked **A** and **B.** The distances are marked on the graphs of velocity, V against time, t from their start as shown below.

The School Nurse was among the spectators and demanded that the boys put on helmets which had a caution on the inside for their safety in case of a fall.





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Another competition involved selecting a participant in a darts game. Two girls Jane and Sofia trained by firing darts at a target $2m$ high above the horizontal. Jane released a dart at 45^0 to the horizontal at $16ms^{-1}$ and Sofia at 30^0 to the horizontal at the same speed like Jane. The target was located $3m$ from the throwing point.

The Coach kept wondering why darts were designed with flight (wings). One student told him that the streamlined wings (flights) at the tail end of darts is to provide stability and produce a more lift force and that this effect has many real life applications which were disclosed.

The team required a physics student to provide technical assistance.

Hint:

An average speed not less than 6.0 ms^{-1} was needed for one to qualify.

Acceleration due to gravity is $9.81ms^{-2}$

Task:

You have been chosen:

(a) Explain motion of the boys and the demand made by the nurse hence select the boy and the girl that will able to represent the school in competitions.
(b) Explain any two of the real – life applications disclosed to the coach.

ITEM 2

Cancer in Uganda is a major, increasing public health crisis. One of the Privately owned Hospital wants to start a cancer treatment department. The proposed cost effective method is the use of an efficient Cobalt ($^{60}_{27}Co$) teletherapy machine which utilizes gamma rays from radioactive cobalt decay to destroy tumors. The radioactive isotope $^{60}_{27}Co$ decays to Nickel (Ni) producing a beta particle and the gamma rays. The project manager sat with some health workers who suggested that before the machine is bought, there should be a clear understanding of the stability of atoms using binding energy per nucleon graph, an equation showing the decay, a determined rate of disintegration and the amount of energy the machine can possibly produce un the process.
The available reference data indicate that an efficient teletherapy machine exploits $20g$ of the Cobalt at a rate of disintegration not more than $9x10^{14}Bq$ to produce at least $5.5x10^{20}MeV$ energy.

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They told the manager to involve a student of Senior Six advanced level physics for help if the best solutions were to be obtained.
The project manager was told that the student should help in describing the structure and operation of a device which uses particles of same properties like the beta particle to produce radiations for use in scanning whether cancer patients who always complain of painful bones have fractures.

Hint:

Half life of $^{60}_{27}Co$ *is* $5.27 years$
Mass of $^{60}_{27}Co = 59.93381U$
Mass of $Ni = 59.93079U$
Mass of an electron $= 0.0005U$ where $1U = 931MeV$

Task:

You have been selected for help
(a) analyze the given data to address the concerns raised by health workers
(b) Explain how the device suggested by the manager can work to serve the desired purpose

SECTION B (PART I)

ITEM 3

Students in a certain secondary school complained of too much heat in the classroom.
They claimed that this could be due to much light which entered through the door which would remain open when one entered or the quality of the newly installed windows.
The head teacher convened a meeting with the class and invited some S.6 physics students, their teacher of physics and the supplier of the windows to address the issues raised.
The supplier said that the windows supplied are of good quality and each window had a double glazed glass consisting of two Sheets of glass $5mm$ thick separated by a layer of air $10mm$ thick. The cross sectional area of the window was $2m^2$ and absorbed 5% of radiation from the sun of solar constant $1100Wm^{-2}$. He also said that the thermal conductivity of the used materials were $0.025Wm^{-1}K^{-1}$ for air and $0.8Wm^{-1}K^{-1}$. He said that all this ensures that even when the outside surface

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of the window was at 35°C, the inside temperature in the classroom would be maintained at most 20°C which is good enough for the conducive atmosphere. The physics teacher raised doubt on whether said inside temperature in the classroom and the thermal conductivity of the glass are accurate. The teacher requested one of the physics students to obtain a small piece of glass material similar to that used in the windows and use it to establish whether it has the accurate thermal conductivity as compared to the standard value known from the market and also establish whether the inner class room temperature was at most 20°C.
The teacher also requested the same student to explain how the outer surface of the window can be made in order to reduce the amount of energy absorbed from light falling on it.

The head teacher remembered that there's a simple device which when installed on the door, it enables it to slowly close automatically after it is opened.

Task:

You have been selected by the physics teacher,

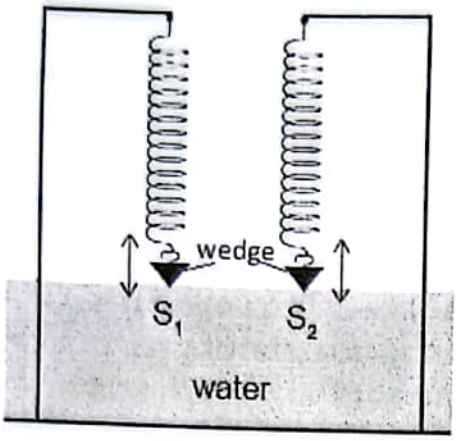
(a) analyze the data to address the requests put forward to calm down the class members about window quality.

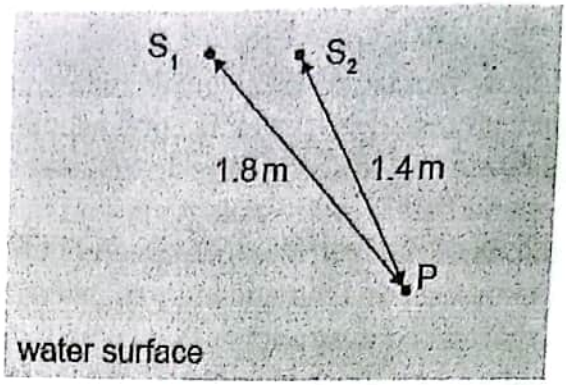
(b) Assess the information given to explain is mode of operation of the said device.

ITEM 4

Energy is a fundamental property of nature that cannot be created or destroyed, only transformed from one form to another for various life applications. A group of S.6 Physics students decided to perform some experiments to enable them understand concepts on energy better. Okello, one of the group members used two identical wedges attached to two identical vertical springs. The wedges touch a water surface at S1 and S2. As the wedges oscillate, they create travelling waves on the water surface. The wedges oscillate in phase with the same amplitude. P is a point on the surface of the water.

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The mass of each wedge is 0.200 kg and the spring constant is 504 $N\,m^{-1}$. Damping is negligible. The speed of the water waves is 1.6 $m\,s^{-1}$. Okello was tasked by the group to find the frequency of the water wave and make them understand the observations made at point P and slightly after.

The head teacher also had a newly purchased a 2.352kW power Kettle made of a material of heat capacity of 420 JK^{-1} and he was assured by the shop owner that the kettle was so genuine and can boil water within 6 $minutes$. The head teacher gave it to the group to test whether it was truly genuine. They heated 2 $litres$ of water initially at 20^0C until it boiled at 100^0. Okello observed that the steam produced had energy and was almost lifting off the cover. The discussion leader asked the group to heat the water for extra 4 $minutes$ and establish the work done by the water against a constant atmospheric pressure of $1.0x\ 10^5Pa$. The head teacher came back and asked them to give the molecular account of all the heating processes.

Rose entered the room and found a great science group. She requested them to recommend a device which can help her vision. She was always complaining of her failure to see far writings clearly.

HINT

Density of water is $1000kgm^{-3}$ while that of steam is $0.6272kgm^{-3}$.
Specific heat capacity of water is $4200Jkg^{-1}K^{-1}$
Specific latent heat of vaporization of steam is $2.25x10^6Jkg^{-1}$

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Task

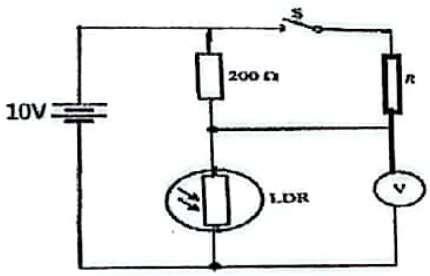
As a student of physics;

(a) Analyze the scenario to address the concerns raised in the discussion group.
(b) Recommend the device for Rose and make her understand how it will be of help.

ITEM 5

Automatic light systems in farming are often powered by off-grid sources. They have sensors and are commonly used for **livestock management, greenhouse horticulture,** and **farm security.** These systems help optimize animal growth, regulate crop photoperiods, and reduce energy costs.

Uganda **Young Tech LTD is** a company that specializes in automated garden lighting have been contacted by Musa a famous farmer to check the effectiveness of his new sensor system to enable it behave consistently whether it's under the faint glow of moonlight or the bright glare of the sun according to the circuit diagram below.



V represents a control unit that operates at a specific voltage, V_m, The resistor **R** has been damaged and needs replacement. It is known that **the Sensors:** The Light-Dependent Resistor (LDR) is the heart of the system. Its resistance is $\mathbf{800\Omega}$ in moonlight and drops to $\mathbf{160\Omega}$ in daylight. With the override switch S open, the control unit which operates perfectly well when the LDR is exposed to moonlight and when the override switch S is closed, the control unit still works perfectly when the LDR is exposed to daylight but V_m value remains unchanged . Ndugu works for Uganda **Young Tech LTD** and was sent by the company to fix the system. He needs to establish if $V_m > 9V$ as required for normal operation of the control unit of the lighting system and the value of the resistance **R** to be bought

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Musa could not let Ndugu return before checking his water pump system because its efficiency was getting weak. Ndugu found out that its motor contains a coil with $500\ turns$ and when running, its magnetic flux changes from $0.05Wb\ to\ 0.02Wb\ in\ 0.2s$. This coil further worked as an inductor of inductance $0.5H$ and resistance 30Ω connected in series with a start-up capacitor of capacitance $1000\mu F$ to provide an initial powerful jolt of energy for the motor coil to overcome resting inertia and balance motor current. It is connected to a $200V\ (RMS)$ supply alternating at $\frac{50}{\pi}\ Hz$. Ndugu wants to check if this coil can withstand at least $70V$ of induced e.m.f and if the system can supply the necessary $8A$ peak current required for efficient use or else the pump be replaced. Musa however asked Ndugu to tell him whether the Motor was A.C or D.C. Ndugu said that this Motor should be A.C but Musa insisted that he should be made to understand how the D.C Motor Works for him to make a clear distinction of what was being used for his pump.

Jane is Musa's farm manager and she advised Musa to procure a device which can boost the a.c voltage supply from 200V to at least 240V for the efficiency of the Pump to improve.

Task:

As a student of physics, you have been approached for help

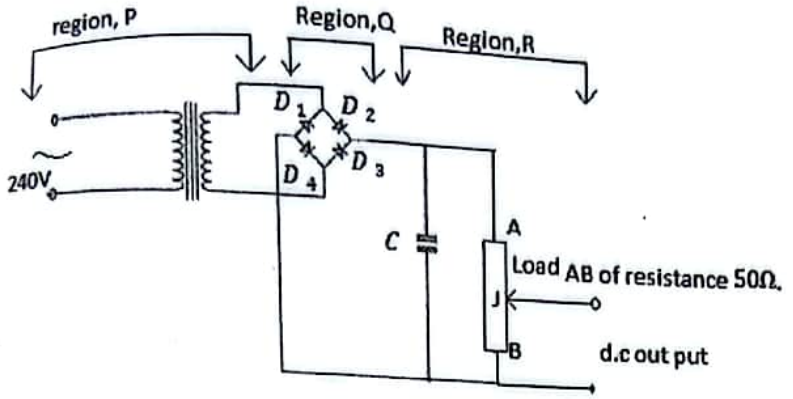
(a) provide the solutions needed by Ndugu to satisfy Musa's demands

(b) assess the design and operation of the suggested device to serve its purpose.

ITEM 6

Most electronic devices which use semiconductors, microprocessors and batteries require d.c voltages converted from a.c source. Okot bought a smart phone but he got challenges with its charger. A typical set –up in figure 5 below is a circuit which is used to charge his phone using a $240V$ a.c voltage source. The charger not working is attributed to various factors but the major ones zero down to charging levels of the capacitor and the value of output voltage.

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The set-up used in region P has primary to secondary turn ratio of $8:1$ and the load AB is of length $100cm$ with the potential divider tapped at point J where $AJ = 30cm$. The capacitor, C used has plate dimensions $10cm$ *by* $10cm$ separated by a dielectric material of constant 3 and thickness $6mm$. The required capacitor should be able to store charge not less than $3.0nC$ and the best charging system should have a power output not less than $5W$.

Okot was told by his visitor that set-up in region P has many applications in everyday life devices.

HINT:

Permittivity of air $\varepsilon_o = 8.85x10^{-12}Fm^{-1}$

Task

You have been contacted by Okot for assistance,

(a) Explain actions in regions P, Q and R. Hence, establish the problem of the smart phone charger.
(b) Describe the structure and operation of a device which uses the principle of the set up in region P.

END

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